Multidimensional Mean Recap

Don't mix up the letters (once more)!

Remember, when we talk about the **population** (all data points), the mean is denoted by the Greek letter μ, and we divide by the total number of elements N, and the standard deviation is denoted by the Greek letter σ. However, when we work with a **sample**, we use $\bar{x}$ as a name for the mean, we divide by n, and our standard deviation will be called s (and in this case, don't forget, we divide by $(n-1)$ in order to calculate the estimate).

In vector form, when we pack everything into arrays for more efficient calculation, we obtain the following:

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}$$ (the **population** mean vector), and

$$\bar{\boldsymbol{X}} = \begin{pmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \bar{X}_3 \end{pmatrix}$$ (the **sample** mean vector).

Attention!

In the previous video, at 0'56, since we estimate the population **mean** through a sample, and not the **variance** or **standard deviation**, we divide by n ($(n-1)$ is necessary only for the variance or standard deviation).

As earlier in the course, the notion of **bias** refers to the fact that if you try and estimate information about the **population** by doing calculation only on a **sample**, your calculation may not lead to an approximation of the desired value. Hence this distinction that you need to remember:

- for the **mean**, as seen in the previous video, the estimate is **unbiased**, which means dividing by n will give you the right result;
- but, as mentioned already, if you are trying to estimate the **variance** or **standard deviation**, you have to use Bessel's correction, that is, you have to divide by $(n-1)$ instead of only by n, which you could do if you had all the elements of your population available.